Trøndelagkonkurranse 2026 - Level I

Third Round - October 31, 2026

1st Part

Choose and solve only one of the following problems:

Problem 1

Find all triplets (a, b, c) of natural numbers such that $a < b < c$ and

$$\frac{1}{a} + \frac{1}{b} + \frac{1}{c} \in \mathbb{N}.$$

Problem 2

A natural number $n > 1$ is called *agreeable* if an equilateral triangle T that can be divided into n smaller equilateral triangles such that:

- The sides of the smaller triangles are parallel to the sides of T.
- The smaller triangles do not overlap inside T. Vertices and edges can touch each other.
- Every portion of T is covered by a smaller triangle.

Show that 5 is not agreeable.